PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

AUTOMATED RESOURCE MANAGEMENT SYSTEM

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The advent of advanced automation technologies has the potential. The need of more efficient tool to manage inventory and advancement of automation technologies pushes the emerging of Automated Resource Management Systems in various forms over several decades, evolving from 1960s manufacturing inventory control into the sophisticated, AI-driven platforms used today. Inventory management is critical to running any business, especially in auto supply businesses,where maintaining an optimal stock level directly impacts profitability and customer satisfaction. Inventory management involves tracking, ordering, storing, and selling a company's stock. In the context of auto
supply businesses, effective inventory management ensures that essential parts and products are available when needed while avoiding the high costs associated with excess inventory or stockouts. Moreover, the recommendation to adopt a "just-in-time" (JIT) inventory system is insightful, for it can help reduce costs by minimizing the amount of inventory held at any given time, lowering storage costs, and reducing the risk of overstocking. This approach ensures that businesses only order inventory as needed, improving cash flow and preventing excess inventory from becoming obsolete or taking up valuable space.

Because of this people are using web-based applications a lot. These applications are used by institutions and organizations. They make it easy for people to talk to each other, store information and manage data. For example schools using these applications. In many organizations, they still use traditional way when it comes to managing equipment and resources. They do not use web-based applications for this. Managing equipment and resources is still done by hand. Due to the manual process, problems often arise such as improper recording of information, data loss, slow processing of transactions, and manual data entry leads to version control issues, while disconnected systems create information silos that hinder strategic decision-making. This is the cases, for organizations that use mobile and web-based applications. Such problems can affect the organization and effectiveness of an institution in managing their resources.

To address these challenges, an Automated Resource Management System can be used that will provide a more systematic way of monitoring, recording, and managing equipment and resources. It offers a lifeline, providing surer footing, a more navigable path, streamline inventory tracking, dramatically improve accuracy, replaces manual spreadsheets with real-time visibility across multiple projects, unified workflows, and provide the firepower that allows companies to respond swiftly to market changes based on deeper insights into consumer behavior and trends. The aim of this study is to develop an automated system that will help facilitate and improve resource management.

1.2 Statement of the Problem

Despite the availability of various technologies that can be used in information management, many organizations will face problems related to manual resource management systems.

This study is going to find out if the Automated Resource Management System really works in making resource management and how the organization runs better. The Automated Resource Management System is what we are looking at. We want to know some things about the Automated Resource Management System.

1. Does the Automated Resource Management System make a difference in how resources are given out and used?

2. Does the Automated Resource Management System make inventory monitoring, control and management better?

3. Does the Automated Resource Management System make tracking and reporting of resources accurate?

4. Does the Automated Resource Management System help reduce delays, mistakes and waste of resources in what the organization do?

5. How better does the Automated Resource Management System make the allocation and use of resources that are available?

6.How well does the Automated Resource Management System manage inventory and the resources of the organization?

7. What kind of effect does the Automated Resource Management System have on how accurate and reliable tracking and reporting of resources are?

8. How much does the Automated Resource Management System help reduce mistakes, delays and waste of resources, in operations?

9. What kind of impact does the Automated Resource Management System have on how productive the organization's how well it performs overall?

Challenges with managing inventory inaccurate inventory tracking, having an excess of unsold merchandise, and insufficient stock to fulfill client needs are the three main issues with inventory management.

• Inadequate or outdated inventory management techniques
• Demand fluctuations caused by customers' changing desires and needs
• Difficulty locating certain commodities in a warehouse
• Depending on the company's field and employer, there are four main kinds of inventory that they may oversee: raw materials, work in progress, finished goods, maintenance, repair, and operations (MRO/Repair) goods.

[redacted]

1.3 Objectives of the Study

1.3.1 General Objective

This study aims to design and develop an Automated Resource Management System that can help in more efficient, orderly, and systematic management of equipment and resources within an organization and to develop an effective system that will streamline the client's business operations and increase efficiency.

1.3.2 Specific Objectives

To achieve the general objective, this study aims to:

1.To design a system that integrates sales tracking and inventory management with proper categorization and maintaining the data quality during the transaction.

2.To develop a user-friendly web-based system that incorporates a Point of Sale (POS) functionality for efficient inventory management, facilitating faster transactions.

3.To incorporate predictive analytics to compensate for the customer's demand for stocks and prevent overstocking of particular product supplies.

4.To test and evaluate the functionality and accessibility of the proposed system.

1.4 Scope and Delimitation

The objective of this project was to create an Automated Resource Management System that could be used to monitor and control resources/equipment at the institution. The system will allow users to login/register, track/monitor resources, store data in a database, and access an administrative dashboard for system administration. The project will only produce a prototype of the Automated Resource Management System for academic purposes; it will not be used in actual production or integrated with other external systems at any other organization.The data collected for the analysis may or may not be the right time for the analysis. The analyzed data may or may provide the accurate results for decision making.

1.5 Significance of the Study

This study is expected to be useful to the following:

Users – The system will help them more easily access and monitor the resources they use.

Administrators – It will provide a more organized and systematic way of managing and monitoring organizational resources.

Organizations – It will improve the efficiency of resource management and reduce errors in recording information.

BSIT Students – It can serve as an example or reference in developing systems related to automation and database systems.

Future Researchers – This study can be used as a basis for further research on resource management systems.

1.6 Definition of Terms

Application Development – The process of creating and designing a software application from planning to implementation.

Automated System – A system that uses technology to automatically perform tasks with minimal human intervention.

Resource Management – The process of organizing, monitoring, and managing equipment or resources within an organization.

Database – An organized collection of information used to facilitate the storage, retrieval, and management of data.

Administrator – The person responsible for the administration and control of a system.

Monitoring System – A mechanism used to monitor and evaluate the status or condition of resources.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

According to a study by Smith and Anderson (2021), the use of automated systems in information management helps to improve the efficiency and accuracy of processes within an organization. Their study showed that digital systems can speed up data processing and reduce errors caused by manual operations.

On the other hand, Johnson et al. (2020) emphasized that the use of intelligent systems and recommendation technologies helps in improving the user experience and user trust in a system.Highly dependent on its inventory management. Meeting consumer demand is a critical business function, and inventory is the glue that holds the purchasing, supply, and production processes together. The advantages of having inventory, such as improved customer service and protection from supply risks, are often outweighed by the significant costs of having too much inventory. Not only that, but new research has demonstrated that ineffective inventory management has far-reaching consequences that impact the performance and market value of both firms in the long run.

A study by Nakamura and Lee (2018) focused on the use of Just-In-Time (JIT) inventory systems in Japan. The research revealed that JIT helps companies minimize inventory holding costs, improve production efficiency, and strengthen supplier coordination. By maintaining only the necessary level of stock at the right time, businesses were able to reduce waste and optimize their supply chain processes. The study concluded that JIT is an effective inventory strategy widely adopted in global industries to improve operational efficiency.

The importance of inventory management and control in improving operational efficiency for companies in different industries is explored in this study. Efficient inventory management is crucial for companies that want to stay competitive and make money in today's market because of how globalization and technology are changing the dynamics of the industry. A company's success or failure is highly dependent on its inventory management. Meeting consumer demand is a critical business function, and inventory is the glue that holds the purchasing, supply, and production processes together. The advantages of having inventory, such as improved customer service and protection from supply risks, are often outweighed by the significant costs of having too much inventory. Not only that, but new research has demonstrated that ineffective inventory management has far-reaching consequences that impact the performance and market value of both firms (Aatif Latif Shaikh, (April 2024).

Also, several African business units and firms have failed, and ineffective inventory management contributes to these failures (Orobia et al., 2020). Holding too much inventory resulted in more capital being tied up, which caused deterioration, obsolescence, and damage. Conversely, a lack of inventory is linked to sales interruption brought on by stockouts, poor customer service, and underused machinery and equipment (Dong & Su, 2010; Gul et al., 2013; Karim et al., 2017; Orobia et al., 2020). In connection with this, their investigations examined whether inventory management influences the link between managerial skill and profitability.

Meanwhile, Malaysian SMEs struggle with irregular inventory, inaccurate estimates, slow customer response times, and improper accounting recording processes that lead to poor performance (Mohamad et al., 2016). Similarly, Abdulrasheed et al. (2013) noted that poor performance from ineffective inventory management solutions causes firms to struggle with irregular delivery, decreasing consumer productive capacity, and excessive production costs. Since there are not enough materials to satisfy the unexpected rise in consumer demand, there needs to be more profits to cover costs and returns on equity; there are material wastes and thefts brought on by surplus stock, and most businesses have communication chains that are not active. Thus, inventory management has become essential for managers in charge of production. Therefore, the focus of their research is on how inventory management affects the profitability of SMEs.

Effective inventory management in warehouses is presently a top priority for Indian firms. Operation managers must address concerns, including reducing inventory carrying costs and product availability (Panigrahi et al., 2021). The results of their research are used to establish how effective inventory management strategies affect steel manufacturing businesses' operational performances. Their study focuses on how inventory automation and inventory distribution turnover approaches may give large steel manufacturing enterprises a competitive edge and boost operational performance. The same problem is experienced by most Nepalese public manufacturing enterprises, resulting in massive losses due to inefficient inventory management (Risal & Acharya, 2021).

Eliminating future hurdles would satisfy public manufacturing enterprises. Inventory management is a broad topic that has received little attention in the Nepalese context (Risal & Acharya, 2021; Sunday & Joseph, 2017). Thus, their studies analyzed public enterprises' inventory management and profitability.In Saudi Arabia, inventory errors significantly impact a company’s financial reports. Most errors result from poor inventory data management, but they can also be brought on by staff needing to count the goods physically. A proper inventory control system must be implemented to minimize costs while maximizing profit. However, many issues still arise, such as a lack of consistency due to no standard operating procedures, a fixed management system, and material discrepancies (Althaqafi, 2020). Therefore, the study investigated how inventory management affects financial performance.

This study defines inventory management, operational efficiency, and financial performance with the problems identified by past studies. Operational efficiency refers to arranging all production units within a company to guarantee that they operate harmoniously to accomplish critical business objectives (Panigrahi et al., 2021). Additionally, Yu et al. (2018) clarify that operational efficiency refers to the firm’s capacity to enhance results by establishing regular routines and standard operating procedures across functional domains. This study defines operational efficiency as evaluating how well a company uses its resources to provide the greatest results. A separate study conducted by Kwon and Lee (2019) and Panigrahi et al. (2021) concluded that different factors have an impact on operational efficiencies, such as Research and Development (R&D) and Inventory Automation Practices (IAP). The studies concluded that these factors (R&D and IAP) significantly positively affect operational efficiencies and influence a firm’s operational efficiency and overall performance.

Effective inventory management enables businesses to achieve higher levels of customer service, lower costs, and improved operational efficiency (Chase et al., 2021). Realizing the expenses of effective inventory management is still essential for Micro, Small, and Medium-Sized Enterprises (MSMEs) in the Philippines to maximize profit. After reviewing the related literature, there are few studies and insufficient data on the impact of inventory management on operational efficiency and financial performance (Elsayed, 2015; Kljenak et al., 2013; Shockley & Turner, 2015).

Inventory management is crucial and frequently consumes a significant portion of an organization’s operational budget; thus, it has to be closely regulated. It has been acknowledged that keeping accurate inventory records and using efficient inventory control strategies are crucial. It is also found that state firms adopted technology less readily than other industries in inventory management, which may impact how well they manage their inventories (Gatari et al., 2022). Similarly, Khan and Siddiqui (2019) assert that effectiveinventory management does not treat every product similarly; instead, it utilizes control and analytical techniques by the relative economic relevance of each good. In addition, proper inventory planning and control is one of the most critical aspects of a company that requires skill and ability to increase overall performance. Inventory planning and management benefits include improved contract compliance, lower procurement costs, better worker engagement, and lower handling costs (Namusonge et al., 2015).

In addition, Islam et al. (2019) examined the elements contributing to poor inventory management in SMEs. They discovered that the lack of a cohesive firm information system and qualified human resources impede effective inventory management. An integrated information system is essential for providing management with real-time information. Furthermore, it enhances departmental cooperation. To enable the creation of an integrated information system, qualified human resources are needed. Employers must continue to train and retain their workers. By improving quality, saving time, and reducing prices, efficient inventory management may distinguish the business from competitors. From the consumers' perspective, due to daily price increases, society would face economic backwardness due to high inflation, resulting in product shortages and consumer hoarding (Anagaw, 2023). Businesses must lower the number of inventories in the supply chain to rationalize storage costs and maintain them (Hugos, 2018). A robust inventory management system's advantages include efficient stock storage, simple warehouse product access, elimination out-of-date inventory, and a general reduction in stock expenditures (Mulandi & Ismail, 2019).Lastly, inventory management is essential for a firm to succeed, and numerous strategies have been devised to maximize stock levels. Some of the inventory management techniques used by businesses are Just-in-time (JIT), First in, First out (FIFO), ABC analysis, Economic Order Quantity (EOQ), and Vendor-Managed Inventory (VMI). For retailers, Just-In-Time (JIT) is a technique that involves receiving inventory just before selling it rather than keeping it on hand for weeks or months until the business needs it (Cvetkovic, 2022). JIT stresses a lean manufacturing system and reduces inventory waste and handling costs (Patterson et al., 2010).

The utilization of the First-In-First-Out (FIFO) strategy is prevalent in inventory management as it guarantees the consumption or sale of the oldest inventory before newer ones. Adopting this method allows companies to mitigate the potential risks associated with inventory becoming obsolete or spoiled while enhancing the accuracy of their financial reporting. FIFO facilitates preserving product freshness, reducing carrying costs, and providing more precise inventory valuation in financial statements (Smith & Johnson, 2019). In contrast, ABC analysis made effective inventory control possible by categorizing inventory based on significance and assigning priority levels to goods (Kumar & Garg, 2017). The Economic Order Quantity (EOQ) approach determines the right number of orders to place to reduce overall inventory expenses (Wang et al., 2017). Vendor Managed Inventory (VMI) is another technique where the supplier manages inventory levels for the buyer, allowing for a more efficient supply chain (Prajogo & Olhager, 2012).Understanding the benefits and limitations of these inventory management techniques is essential for businesses to choose the most appropriate technique for their specific needs, as it significantly impacts a firm's profitability and ability to meet customer demand (Smith & Johnson, 2019).

Thus, having a thorough grasp of operational efficiency and the variables that affect it may aid firms in identifying areas for development and putting into practice efficient tactics to increase performance.In addition, operational efficiency is a critical factor in the success of businesses across industries. Businesses that emphasized operational efficiency often had better levels of productivity, lower costs, and higher customer satisfaction (Karia et al., 2018). Businesses must be completely aware of their processes, resources, and goals to achieve and sustain operational efficiency. This entails choosing and tracking key performance indicators (KPIs) such as cycle time, throughput, and inventory turnover, as well as routine monitoring and analyzing data to spot areas for improvement. Moreover, successful teamwork, ongoing improvement, and good communication are essential for establishing and maintaining operational efficiency (Sohal, 2016). Overall, firms may gain a considerable competitive edge by realizing the value of operational efficiency and putting out effective plans to achieve it.

To maintain track of its raw materials, stock, work-in-progress, and finished items, every business must have efficient inventory management (Rasool, Hussain, Ullah & Shehzadi, 2024). Moreover, Mamuda and Adamu (2023) contended that inventory management is essential to enhancing the effectiveness and efficiency with which businesses handle their stock. To satisfy consumer needs, Tadayonrad and Ndiaye (2023) emphasize the critical role that correct inventory information plays. They also stress the clear link between improved customer satisfaction and efficient inventory management. Accurate inventory control is essential for companies as it affects not just financial statements but also customer contentment and overall productivity. Inventory management errors may have crippling financial effects and damage a company’s reputation by causing stockouts and overstocking, among other problems (Groenewald & Kilag, 2024). Alam, Thakur, and Islam (2024) state that the biggest obstacle to managing inventory at an optimal level is the difficulty in forecasting demand and aiming customers’ expectations of product availability in the market. One challenge in inventory management is keeping the supply and demand for inventory in balance. In an ideal world, a company wouldn’t lose sales due to stockouts and would have adequate inventory to match client demand. However, because carrying inventory is expensive, the company does not want to keep an excessive amount of goods on hand. The methods for controlling, storing, and keeping track of inventory products are the three primary components of inventory management, according to Zoho Corporation (2024). Managing storage space effectively is a daunting challenge. Utilizing inventory management platforms to plan and construct storage areas gives you more control over when new goods are delivered, which is a crucial factor in determining available space. While ordering to delivery is just one part of the extensive cycle that is inventory management, storage is one of the most important. Regretfully, many of the biggest inefficiencies are also located here (Vyas, 2023). According to Jenkins (2022), new challenges for warehouse design and storage arise when considering biodegradable packaging or doing away with packaging entirely to decrease waste. It can also include purchasing new machinery or a reduced shelf life for some things. One issue with the supply chain may be the loss of inventory as a result of theft, damage, or spoiling. Problem areas must be identified, monitored, and measured.

According to Anshur, Ahmed, and Dhodi (2018), improper inventory tracking increases the likelihood of inventory issues, which also have an impact on wastefulness and additional costs overall. It is laborious,redundant, and prone to mistakes to use manual inventory tracking techniques across several applications and spreadsheets (Vyas, 2023). Accurate monitoring is crucial in the cutthroat industry of today. The use of manual processes continues to be a barrier; many companies still enter data by hand using spreadsheets or antiquated software, which can result in errors (Jenkins, 2022). Periodic and Perpetual systems are two types of inventory control systems. Various methods are used in inventory control to keep an eye on the movement of stocks in a warehouse. Existing stock at a warehouse is managed via inventory control (Estrellas, 2024). Inventory and warehouse managers utilize a variety of inventory management control approaches, including inventory forecasting, inventory reorder points, information technology, and inventory turnover, to monitor and enhance organizational performance (Anshur et al., 2018). There are several approaches to inventory management, and each has advantages and disadvantages of its own. Managers should be sure to take into account the kind of product they offer, the size of the company, the total budget, and the degree of precision required to operate a successful supply chain when selecting the best strategies (Tarver & Aditham, 2023). It is a function that is very vital and significant to the performance of any kind of organization (Hayes, 2024; Porzuczek, 2022).

Munyaka and Yadavalli (2022) stated that deterministic inventory theory bases inventory operations on a known (certain) demand. A deterministic model yields the same result due to certainty about the factors, conditions, and parameters involved, which are stated explicitly at the beginning. One of the parameters is demand. Antic, Djordjevic, and Lesec (2022) state that deterministic demand is expressed as a monthly sales forecast for each product. The deterministic demand model seeks to minimize the total costs associated with production time, setup time, and overtime, as well as inventory-related costs like ordering costs, carrying costs, and stock-out costs (overstocks and shortages).According to David (2019), deterministic theory is the inventory theory that most suppliers or retailers use. In essence, this is maintaining inventory levels and placing further orders as needed. When shipment times and client orders can be predicted, this hypothesis performs well. You need to have enough inventory, for instance, if it takes a week for things to ship to astore. If not, you will have to decline orders from consumers when the item runs out of stock.

According to Dan C. (2014), E-commerce can provide substantial benefits even to small enterprises through improved efficiencies and raised revenue. It enables a new way of working to emerge as business face the future and embrace the new economy. E-commerce enables small business entrepreneurs to gain access to better quality information, thus, empowers them to be informed about their decision in their business. Most importantly, E-commerce can give a competitive advantage. It can help strengthen market position and open new business opportunities with its potential to improve profits. Likewise, the statement provided by KPMG International (2017) describes the capabilities of E-commerce when it comes to accessibility. Actually, even businesses without physical location could operate because of this technology.

According to Sharma G. (2015), E-commerce is considered an excellent alternative for individuals and companies to reach new customers. Service quality delivery through Internet is an essential strategy to success, more 93 important than price and web presence. The E-commerce Website has been identified as having a significant impact on business activities in solving the geographical problem. A number of performance problems have been observed for E-commerce Websites, and much work has gone into characterizing the performance of webservers and Internet application.

According to Pandey D. and Agarwal V. (2014), global presence has been yet another versatile and important advantage of E-commerce, which enables accessibility of a very wide range of products and services virtually, anywhere in the world, even in the remote areas. The feature of global presence has a boosting effect in the entire gamut of international trade, which in turn pose a subsequent impact in the enhancement of an even greater level of global business presence of products and services offered. Hence, this relationship is vice-versa. Technology driven E-commerce can provide automated continuous business transaction round-the-clock. Round-the-clock busi-ness operation and the feature of global market presence have enabled the mankind to exceed the physical limitation of Intraday market operations and extended it into a 24/7 process.

However, Inventory optimization and management have a significantly large impact on the retail industry. Every retailer aims to optimize their return on investment, and in order to do so, they must make sure that they are informed of the changes and that their inventory investments are kept at the best, or optimal, level, which is a difficult objective to achieve. Retailers encounter significant difficulties managing and storing their inventory. In a related study by (Stanelyte, 2021), it was determined how the demand of the consumer drives the entire phenomenon of nventory in the retail industry. Since products in the retail sector tends to sell quickly, it is crucial to predict client demand precisely in order to develop the ideal model for inventory replenishment. In addition, in-depth data analytics systems that may be used to predict future client needs based on purchase history data are only one of the tools available to retailers to help them satisfy customer demand. Other factors, including the price that customers pay for products, the expense of maintaining inventory, and the management of product loss, can still lead to poor demand management performance even when inventory managed properly. Retailers can utilize a variety of inventory management techniques to assist their stores satisfy customer demand. The local market environment and the retailer's product line determine the best inventory levels for a certain store. (Chawla, et.al., 2019).

That is why to efficiently track inventory, each online business needs a different approach to inventory management (Lopienski, 2021). Furthermore, because it has complete access to real-time data, implementing an inventory management system is a proven approach to distribute inventory efficiently as the supply chain expands (Lopienski, 2021). Inventory management affects how well a company runs its operations, treats its customers, and boost revenues.In order to meet demand, inventory optimization seeks to cost-effectively order the appropriate amounts of the right products. This is accomplished by incorporating supply and demand volatility into stock rule and inventory parameter adjustments. Inventory optimization seeks to achieve optimal availability while lowering inventory costs and reducing the risk of expensive surplus or obsolete stock. It does this by balancing service level (product availability) goals with inventory investment restrictions (Drakeley, 2023).

Integrating digital technologies to reduce errors, delays,and administrative burdens, to enhance cost-efficiency and responsiveness. Likewise, Abubakar et al. (2019) emphasized that the lack of integrated inventory systems in educational institutions leads to higher operational risks and limits the institutions' capacity for strategic asset planning. In addition, the integration of data mining techniques into inventory management has been shown to enhance forecasting accuracy, enabling institutions to prevent shortages and overstocking. This approach also improves monitoring efficiency, speeds up reporting, supports more effective resource allocation (Tungcul & Kummer, 2021). Effective tracking of assets prevents resource loss, redundancy, and inefficiency, enabling organizations to maintain optimal resource allocation.

Traditionally, inventory management practices, such a spreadsheets, are labor-intensive, error-prone, and inefficient. These methods often lead to data inaccuracy, oversight, and internal shrinkage, significantly impacting business operations (Madamidola et al., 2024). In response to these trends, some auto supply businesses are adapting their inventory management practices to better align with their specialization needs. By adopting new inventory management techniques, these businesses can gain greater control over stock levels and streamline their ordering processes, ultimately enhancing efficiency and operational effectiveness.Understanding the significance of physical location in inventory management is crucial for auto supply businesses. Smith and Johnson (2018) emphasized that the location of retail establishments greatly influences inventory turnover rates, demand forecasting accuracy, and transportation costs. Urban locations often have higher demand but may face challenges related to space constraints and traffic congestion (Rahman et al., 2021).

Conversely, rural areas might offer lower operating costs but could struggle with lower customer traffic (Li & Liu, 2012). On the other hand, the longevity of an auto supply business can significantly impact its inventory management practices. Jeong et al. (2019) noted that established businesses often have refined inventory control systems, benefiting from historical sales data for more accurate demand forecasting. Conversely, newer businesses may face challenges such as inventory overstocking or stockouts due to limited historical data (Garcia & Nguyen, 2019). Inventory management is the art and science of maintaining stock levels of a given group of items, incurring the least cost, and being consistent with other relevant targets and objectives set by management (Prempeh, 2015). It is an important part of making all the decisions in handling the inventory in an organization, such as activities to be carried out, policies of inventory management, and procedures in handling the inventory to ensure enough quantity of each item is always kept in the warehouse. (Chan et al., 2017). Effective inventory management practices (EIMP) are essential in the operation of any business. However, implementing such practices faces challenges, notably cost barriers, attitudes, and knowledge of owners/managers (Atnafu et al.; Ahmad & Zabri, 2018). Overcoming these hurdles is vital for sustaining growth and competitiveness in the market. Furthermore, the effectiveness of inventory management practices may directly impact customer satisfaction through ordering decisions, minimizing inventory levels, and eliminating costs (Ahmad et al., 2020; Andersson et al., 2015; Smith & Jones, 2018). While optimizing inventory ordering decisions and minimizing levels are essential for operational efficiency (Orobia et al., 2020), eliminating costs must be balanced to avoid stockouts and maintain customer service levels (Simchi-Levi et al., 2015).

According to (Wang & Hong, 2022), thousands of final products together with extensive types of raw materials and intermediates make large-scale production networks complicated. Traditional inventory models cannot manage the intricate decisions often needed to be made, while traditional simulation methods are too slow for handling the scale effectively. For solving these drawbacks, (Wang & Hong, 2022) proposed a simulation approach which is RNN-inspired: it exploits the computational potential of recurrent neural networks for the purpose of production network structural layouts. This technique proved very fast: by orders, typically by thousands, more rapid compared to conventional simulations and yet will enable an efficient optimization even of the biggest inventory-related problems in the time constraint of real-world applications.

Ain, Kiisler., Olli-Pekka, Hilmola. (2020) explained Abstract Research is based on wholesale and distribution operations of real-life case company, and in this setting, the most critical part of company's supply chain is the inventory replenishment to warehouse (Distribution Center) as well as fulfilling and delivering customers' orders. Different Economic Order Quantity (EOQ)-based models have been considered (Reorder Point, Reorder Point with pipeline on order inventory, and "pulse train"). Simulation system evaluates annual total logistics costs. Results show that in an environment, where local warehouse inventory levels are rather high and replenishment order quantity is rather small, it is important to have frequent shipments divided in suitable intervals. In the simulation model, this could be done e.g. with the use of "pulse train" function or incorporating a pipeline on order inventory in order to decide. The research findings are valid for a small-scale supply chain servicing small and geographically limited markets with clients assuming high customer service levels (e.g. 24-hours lead time). For bigger markets, the cross-docking based supply chain models are worth considering in simulations.Recently, Li et al. developed an affordable AI-assisted machine supervision system for SMEs manufacturing that significantly reduced costs and provides actionable insights and engage SMEs in sustainable manufacturing.

However, sales are not a universally ideal indicator of performance. It is affected by factors such as inflation and currency exchange rates, whereas employment remains relatively stable in this.regard. Moreover, sales do not always precede growth in other areas. In high-technology start-ups and the launch of new activities within established firms, it is often observed that assets and employment may grow before sales are realized (Delmar et al., 2003). On the other hand, employment has been argued to be a moredirect indicator of organizational complexity than sales, making it a preferable metric when the focus is on the managerial implications of growth or performance (Delmar et al., 2003). Studies also show a strong link among sales growth, an expanding customer base, and increased market reach, which collectively drive the need for additional workforce (Storey, 2016). MSMEs with well-structured strategies and a focus on market penetration demonstrate a higher capacity for job creation. Scholars often emphasize employment growth as a vital measure of business performance and its impact on job creation (Siepel and Dejardin, 2020). This measure is particularly relevant to policymakers, who tend to prioritize employment over sales growth in assessing the economic contributions of businesses. A finding supported by Şeker & Saliola (2018), who highlights the remarkable job growth rates among small firms in developing countries. Similarly, Clayton et al. (2013) point out that while small businesses are often regarded as the primary drivers of job growth, older firms also play a substantial role. Although entrepreneurs and managers may not always prioritize employment growth as a direct indicator of success, it remains a critical measure for understanding medium- and long-term business development.

On the other hand, subjective measures focus on qualitative aspects of effectiveness, such as growth, expansion, efficient service delivery, product quality, survival, and competitiveness (Dobbs and Hamilton, 2007). Also, through extensive literature reviews, Ardichvili et al. 2003). However, for this study, it focused on sales and employment growth only.The measurement of MSME business performance can be seen from the increase in sales, which is the increase in the number of purchases made by customers for the goods sold (Agus et al., 2023)). Studies indicate that a range of factors contribute to sales improvements within MSMEs. Access to finance plays a critical role, enabling investments in inventory, expansion, and marketing efforts (Nwosu and Orji, 2016). Strategic marketing skills, particularly in leveraging digital channels, have become vital for reaching wider audiences and promoting products and services (Purwanti et al., 2022). The development of innovative products or services that cater to evolving customer needs can be a significant driver of sales growth (Halila and Rundquist, 2021). Moreover, building strong customer relationships through personalized service and after-sales support leads to enhanced brand loyalty and repeat purchases.

One critical aspect of tangible resources is their contribution to operational excellence. Astudy by Wongwilai (2022) emphasizes the importance of modern and well-maintained physical assets and infrastructure, such as machinery, equipment, facilities, and inventory. These assets enable MSMEs to streamline production processes, reduce operational costs, enhance product quality, and meet customer demands efficiently. Effective utilization of physical assets leads to improved operational efficiency and productivity, thereby contributing to overall performance.Furthermore, the strategic allocation of tangible resources contributes to MSMEs' competitive advantage. Moscare-Balanquit (2021) suggests that MSMEs that strategically invest in tangible assets, optimizing operational costs, and managing supply chains efficiently are better positioned to achieve sustainable competitive edges in the market. Effective resource allocation enables MSMEs to differentiate their offerings, enhance customer value propositions, and respond promptly to market demands, thereby strengthening their competitive positions.One key aspect of optimizing physical capital is ensuring efficient resource utilization. Studies like the one conducted by Mugo et al. (2019) on Kenyan manufacturing MSMEs highlight the importance of capacity utilization. This refers to getting the most out of existing equipment and facilities. By operating at optimal levels, MSMEs can maximize output without compromising on quality. Proper maintenance of physical assets is equally crucial. A study by Salawu et al. (2023) emphasizes how preventive maintenance practices minimize breakdowns and production delays, ensuring smooth operation and maximizing uptime. Strategic investment in physical capital can also propel MSME growth. Upgrading equipment with newer technologies can lead to several benefits. Santanaa et al., (2018) highlight how this can increase efficiency, improve product quality, and even allow MSMEs to expand into new markets with capabilities that cater to evolving customer demands. However, informed decision-making is critical. The same study also emphasizes the importance of careful planning and financial resource consideration to avoid overinvestment in assets that may not be fully utilized or become obsolete too quickly.

Research across various regions reinforces the positive impact of physical capital on MSME performance. Sandu et al., (2022) conducted a study in Kenya that found a strong correlation between effective physical capital management and profitability in small businesses. This highlights the financial benefits of optimizing resource utilization and making strategic investments. Similarly, Mugo et al.'s (2019) study provided evidence of the link between capacity utilization and firm performance, demonstrating how getting the most out of existing equipment translates to better business outcomes.The critical role of financial resources in MSME performance extends beyond just having available cash flow. It also provides the essential capital needed to initiate and sustain business activities, ranging from purchasing inventory and equipment to covering operational expenses and salaries. Adequate financial resources ensure smooth cash flow, enabling MSMEs to meet their immediate financial obligations and seize opportunities for growth (Nkwinika and Akinola 2023).Moreover, the ability of MSMEs to access credit and external financing options is instrumental in fueling expansion, innovation, and investment in new technologies or market opportunities (Bose, 2013). A healthy financial profile with access to credit facilities supports ongoing operations and empowers MSMEs to pursue.strategic initiatives that drive long-term competitiveness and market relevance. Enhanced access to credit enables investment in assets, expansion, and adopting new technologies, facilitating business growth (Adjabeng and Osei, 2022). Studies indicate a positive relationship between financial inclusion and MSME profitability, productivity, and job creation (Nwosu and Orji, 2016; Bruhn et al., 2017). Access to savings accounts allows MSMEs to manage cash flow effectively and build financial resilience during downturn.

This positive impact of employee experience on MSME performance is supported by various research studies. Suroso et al. (2017) found a significant correlation between employee experience and SME business performance. Similarly, Ali Hamid Ibrahim's (2018) research in Libya highlighted the significant contribution of experience to an MSME's success. Egberi's (2019) study in Nigeria further confirmed that employee experience has a positive and measurable effect on an MSME's overall performance. The knowledge held by employees stands as an indispensable asset within 'he framework of human capital for any organization. This knowledge comprises various dimensions, including educational attainment, the culture of knowledge sharing, disciplinary acumen, and accumulated experiences (Isaga, 2015). Employees endowed with profound knowledge play a pivotal role in advancing product or service innovation and elevating overall customer satisfaction levels. Furthermore, their expertise serves as a catalyst for streamlining production processes, thus contributing to cost reduction and enabling efficient resource allocation within the company (Ibrahim, 2018).

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While physical assets are essential, an MSME's true competitive advantage often lies in its intangible resources. These are the non-physical assets that drive innovation, build customer loyalty, and enhance brand reputation. Research in this domain reveals insights into how these intangible resources impact MSMEs' competitiveness and success (Stan et al., 2024). A strong brand reputation, built through effective marketing, quality commitment, and positive customer experience, is a valuable intangible asset. This reputation can be leveraged in global markets, with individual culture playing a significant role (Agarwal and Osiyevskyy, 2021) Organizational culture, particularly in MSMEs, is crucial for driving employee behavior, engagement, and productivity. Corporate governance principles, including fairness, accountability, responsibility, and transparency, can enhance employee commitment, with corporate reputation and organizational trust mediating this relationship (Yeo et al., 2023). The importance of reputation as an intellectual asset for company competitiveness is also highlighted (Krstić et al., 2021).

Building trust among customers is the foundation for growth. Positive brand reputation fosters customer loyalty, leading to repeat business and increased customer lifetime value. In the service sector, brand reputation plays a key role in fostering customer loyalty, with brand trust being a significant factor in this relationship (Gökerik, 2024). Brand credibility further enhances customer loyalty by reducing switching behaviors and increasing word-of-mouth (Tabelessy, 2025). Trust is also a critical factor in relationship marketing, influencing service performance and fostering commitment. Loyal customers become brand advocates, promoting the MSME through positive word-of-mouth. This organic customer acquisition reduces marketing costs while attracting new customers who trust the brand reputation (Gowthaman and Rau, 2010). A strong brand reputation, as evidenced by a positive corporate image, can lead to a significant market-value premium, superior financial performance, and lower cost of capital (Smith et al., 2010). This is due to the ability of strong brands to command higher prices, create a loyal customer base, and maintain brand stability (Hoeffler and Keller, 2003). Furthermore, firms with good reputations are better able to sustain superior profit outcomes over.time (Roberts and Dowling, 2002). This is particularly relevant for MSMEs, as they can benefit from higher profit margins and long-term customer loyalty (Tijani et al., 2023).

The digital age presents a unique opportunity for MSMEs to leverage brand storytelling in building and managing their reputation. Social media platforms and online reviews provide powerful tools to connect with customers, share the MSME's brand story, and showcase its values (Ansarin and Ozuem, 2014). By fostering positive online engagement and addressing customer concerns promptly, MSMEs can cultivate a positive brand image and build trust. Thaichon and Quach (2015) highlight the importance of brand reputation in the competitive game, particularly in stable market segments. They further emphasize the role of brand equity in shaping customer expectations, satisfaction, trust, and value, ultimately influencing customer retention. Keller (2009) provides a practical framework for building and managing strong brands, emphasizing the need to understand consumer brand knowledge structures and the use of online, interactive marketing communications. A positive organizational culture, characterized by support and collaboration, is crucial for fostering employee engagement in MSMEs. Engaged employees, who are committed, passionate, and energetic, contribute to reduced turnover, increased productivity, better customer service, and creative problem-solving (Dickson, 2008). This engagement is particularly important in the manufacturing sector, where it can significantly impact employee retention, productivity, and loyalty (Mohanty et al., 2020). Engaged employees are more motivated, take ownership of their work, and are more likely to go the extra mile for the MSME's success. Furthermore, highly engaged employees can lead to a 147% outperformance of competitors, as well as reduced absenteeism and employee turnover (Elliott, 2023).

2.2 Local Studies

In the Philippine context, According to Dela Cruz and Santos (2022) application-based systems developed in local institutions help improve data management and process automation. Their study also showed the use of modern automation systems automation. Their study also showed the use of modern automation systems. Similarly, Dela Cruz and Santos (2020) examined inventory management practices among small and medium enterprises (SMEs) in the Philippines. The researchers found that effective inventory control helps businesses improve stock monitoring, reduce overstocking and stockouts, and increase profitability. The study emphasized the importance of using inventory management systems, conducting regular audits, and training employees to ensure efficient operations. Overall, the findings showed that proper inventory management significantly enhances business performance in local enterprises.

According to Janice E Carrillo, Michael A Carrilla, (Jan 2024): A company's success or failure is highly dependent on its inventory management. Meeting consumer demand is a critical business function, and inventory is the glue that holds the purchasing, supply, and production processes together. The advantages of having inventory, such as improved customer service and protection from supply risks, are often outweighed by the significant costs of having too much inventory. Not only that, but new research has demonstrated that ineffective inventory management has far-reaching consequences that impact the performance and market value of both firms in the long run. The fundamentals of efficient inventory management are laid forth in this chapter. Along with more contemporary supply chain and e-commerce strategies, we examine traditional methods of inventory management. In addition, we go over some of the technology that really make inventory management a breeze. The chapter concludes with a summary of the most important managerial issues for inventory management.

Therefore, the study investigated how inventory management affects financial performance. Parilla et al. (2022) assert that one of the crucial management dilemmas for various companies in the Philippines is how they manage their inventory. Poor customer service, dwindling resources that might prohibit things from being sold, and unused gear and equipment are all problems with inventory management. Higher rates of inefficiency characterize this, declining productivity, rising operational expenses, bolstered working capital, significant declines in reaching customer service goals, and a precarious bottom line. Poor stock management puts firms in a logistical and financial bind by reducing their efficiency, negatively impacting their financial performance.

2.3 Related Literature

Operational efficiency, customer happiness, and profitability are all impacted by inventory management, which is a fundamental component of supply chain management. This research looks at modern methods of inventory management, specifically at tactics that strike a compromise between operational efficiency and cost-effectiveness. The research's overarching goals are to determine what constitutes optimal inventory management practice, examine how methods like Just-In-Time (JIT), Economic Order Quantity (EOQ), and ABC analysis affect company performance, and draw attention to the ways in which technology might enhance inventory control.

The research assesses the efficacy of inventory management in lowering costs, minimizing stockouts, and optimizing resource allocation through the use of case studies, data analysis, and interviews with industry experts. Automation, realtime inventory tracking, and predictive analytics are highlighted as key components of modern inventory system. Inventory management is the process by which a business keeps track of its stock, including production, ordering, storage, and sales. The business is responsible for managing the warehousing and processing of all materials, from raw materials to finished products. A company's ability to maintain profitability, respond to trends, avoid supply chain management failures, and remain organized is proportional to the quality of its inventory management.

According to Pressman (2019), a proper software engineering process is essential to ensure the quality and reliability of a system. Apart from that, studies also show that the use of automation helps in improving data accuracy, system security, and operational efficiency of organizations.It is critical for supply chain partners to enhance cooperative relationships, decrease inventory, increase forecasting accuracy, and supply chain efficiency through the use of CPFR, a collaborative method for managing inventory in the supply chain. A definition of CPFR, an explanation of the CPFR method, examples of advantages, challenges to implementation, and solutions to these problems are all included in the article(Pei Zhang, A Qian Yuan, (Aug 2014).

According to Dr. Nivetha P , Mr. Abimanyu, (Dec 2024) Businesses cannot afford to waste money, enhance operations, or lose consumers due to ineffective inventory management. Automation, artificial intelligence (AI), and data analysis have replaced antiquated manual processes as the backbone of current commercial inventory systems. Inventory management systems have grown in significance and influence over the years, and this research delves into all of that and more. As an added bonus, it explains how these technologies help businesses remain competitive while also discussing the pros and cons of utilizing them. Management of inventory, company operations, optimization of resources, contemporary business, automation, artificial intelligence (AI), and customer happiness are all keywords.

The inventory tracking system also known as the inventory monitoring system, according to Zoho Corp (2024), assists you in keeping an eye on the whereabouts and movements of your company’s inventory. The majority of inventory tracking systems are a component of inventory management software, such as Zoho Inventory, while there are standalone systems designed for this purpose. On the other hand, according to Nicole(), inventory tracking is the act of keeping track of all the goods and SKUs that you own, are transporting to and from your warehouses, and the amounts that are available everywhere. Maintaining inventory levels is essential for warehouse and order fulfillment processes. Inventory is tracked using two primary methods: digital systems and manual systems.There are two main systems of tracking inventory which are a manual-based system and a digital-based system. The manual bases system involves the use of pen and paper to physically count, trace, and check inventories from time to time be it periodic, perpetual, or otherwise, whereas, the digital-based system deals with the use of software or advanced inventory management systems to count, trace and check inventories from time to time. Adam (2024) stated that organizations typically maintain sophisticated inventory management systems capableof tracking real-time inventory levels popular POS systems like Square, Vend or Lightspeed offer inventory systems that let you do a whole lot more.

Businesses need to manage their inventories well to succeed. By carefully planning and controlling their inventory, organizations may optimize overall operational efficiency and accomplish particular goals by maintaining ideal inventory levels. By focusing on streamlining and standardizing inventory management procedures, it increases the efficiency of the company's inventory (Hänninen, 2024). According to Mamuda and Adamu (2023), SMEs must use efficient inventory management techniques to improve performance and boost their competitiveness. An organization's performance may be attributed to the reduction of operational issues through efficient inventory management. Despite the accepted value of inventory management in organizations, many SMEs still find it difficult to set up a thorough and efficient inventory management system (Hänninen, 2024). Several studies have been carried out by scholars in academic on inventory management, prominent among them are Olaide and Omodero (2023), Mamuda & Adamu (2023), Iliemena, Odukoya, and Aniefor (2022), Chizoba, Clara, and Juliet (2022), Eze & Uchenu (2021), Sonko & Akinlabi (2020), Olowolaju and Mogaji (2020), Folajimi et al. (2020) Ugwu and Nwakoby (2020) Ogidiolu, Akinosun, and Ajakore (2019), etc, all these studies and to the best of the researchers' knowledge, did not investigate the inventory management practices with the dimensions of inventory control techniques, inventory storage system, and inventory tracking system specifically in Lagos state, hence the gap this study fulfills. SMEs who fail to practice inventory control techniques may tend to buy and hold unnecessary goods which might affect their performance. Also, SMEs who do not put adequate storage systems in place may lose customers which would influence sales, profits, customer satisfaction, market share, etc. Finally SMEs who do not track their inventory might lead to stock pilferage, buffer stock position, expiry of products, loss of customers, loss in profit etc.

Inventory is a crucial asset for many organizations, according to Olaide and Omodero (2023), who referenced Gokhale & Kaloji (2018). This is because inventory is often a substantial asset reported in fiscal statements and provides a source of returns through item sales in the near future. Inventory is also defined as "the stock of any item or resource used in an organization" by Okolocha, Anuri, and Anugwu (2022) citing Davis et al. (2003). Inventories are stockpiles of semi-finished, finished, and raw materials that businesses hold to ensure a smooth production process (Otuya and Eginwin, 2017). Kilonzo et al. (2016) define inventory as idle physical products or stock with a high economic value that organizations hold to package,process, or prepare for sale.

While MSMEs in the Philippines have consistently faced challenges such as limited access to finance, technology, and skills, information gaps, and issues with product quality and marketing they have demonstrated resilience by continually generating employment opportunities, adding economic value, and playing a significant role in export activities (Bolido, 2020). The Philippines presents a compelling local casestudy. Mendoza (2015) found out that while Filipino MSMEs perform reasonably well in terms of activity, their profitability presents challenges. This suggests that access to financial services could be a vital element in improving profit margins within this sector. Additionally, Jou (2022) highlights complexities in local policy, noting that financial considerations often play a role on how local councils support MSME development. Moreover, effective financial management practices further amplify the impact of financial resources on MSME performance. This includes prudent budgeting, cash flow management, cost control measures, and investment strategies aligned with business goals (Das and Mahapatra, 2021). MSMEs that employ sound financial management practices are better equipped to weather economic fluctuations, mitigate financial risks, and capitalize on growth opportunities (Kumar, 2013). Furthermore, sound financial management is crucial for product innovation, which in turn enhances MSME performance (Ferine, 2023). A range of studies has consistently highlighted the critical role of financial resources in driving the performance of MSMEs (Omer et al., 2018; Sarkar, 2016). Beck et al. (2014) found a positive correlation between access to finance, strong financial management.practices, and firm performance in developing countries. Similarly, Berger and Udell's (2004) study in the United States demonstrated that access to credit positively impacts small business growth.

While this definition may differ from other authors, the management literature predominantly attributes human capital to prior education and relevant experience (Taboada and Moya, 2014; Harris et al., 2012; Chen et al., 2006). Investing in human capital is crucial for companies as it enables them to bolster competitiveness, productivity, and profitability. Every individual brings a unique set of skills, knowledge, and experiences to the table. By prioritizing human capital investment, companies facilitate the homogenization of knowledge, thereby amplifying financial performance and strengthening competitive advantages (Francisca et al., 2023). Employee experience, particularly prior industry experience, plays a crucial role in MSME success. Employees with a strong background in the relevant sector possess a deeper understanding of industry dynamics, challenges, and best practices. This translates to several key benefits for MSMEs. Firstly, experienced employees can meet company expectations more readily. They require less training and onboarding, and can quickly adapt to existing workflows, contributing meaningfully from the outset (Muda and Rahman, 2016). Secondly, their experience equips them with the necessary skills and knowledge to perform their duties effectively, leading to fewer errors and efficient task completion. Finally, industry-specific experience allows employees to anticipate and address client needs with greater accuracy, boosting customer satisfaction and loyalty (Jayabalan et al., 2020).

2.4 Synthesis of the Reviewed Literature

Inventory management is crucial to a company's success. It can hand may be time-consuming, error prone, and inefficient for SMEs with limited resources. Automation can increase inventory management efficiency and accuracy. Manual inventory management can cause time-consuming inventory counts,difficulty tracking inventory levels, inaccuracies, and greater expenses due to overstocking or stockouts. These may be reduce production, profitability, and customer satisfaction. efficiency, inventory control, overstocking, stockouts, and data-driven decision-making. High implementation costs, technological complexity, and staff training are other drawbacks.

Automatic inventory management solutions benefit SMEs the most, according to research. One researcher found that SMEs using an automated inventory management system had better inventory control, cheaper inventory holding costs, and higher revenues. In a manufacturing SME case study, an automated inventory management system reduced lead According to the study, automated inventory management solutions can benefit SMEs. Firms must assess the costs and advantages of deploying such systems to ensure they meet their needs and goals. Businesses should consider technological complexity and personnel training before deploying automated inventory management solutions.

Localized inventory automation represents a distributed approach to inventory management that leverages technology to optimize inventory levels and decisions at multiple locations throughout the supply chain network. This approach moves beyond traditional centralized inventory management models to create more responsive and flexible systems that can adapt to local market conditions while maintaining overall system optimization. The theoretical foundation for localized automation draws from distributed systems theory, control theory, and optimization methods to create coherent frameworks for managing complex inventory networks. The automation component of localized inventory systems involves implementing intelligent systems that can make inventory decisions with minimal human intervention, using algorithms that incorporate real-time data, predictive insights, and business rules to optimize inventory levels continuously. These systems typically employ optimization algorithms, control systems, and decision support tools that can process large amounts of data and make rapid decisions in response to changing conditions.

The automation enables faster response times and more consistent decision-making compared to manual inventory management processes. Network theory provides important insights into the design and operation of localized inventory systems, particularly regarding the optimal distribution of inventory across multiple locations and the coordination mechanisms necessary to maintain system-wide performance. The design of effective localized inventory networks requires careful consideration of trade-offs between local responsiveness and global optimization, inventory holding costs and service levels, and automation benefits and implementation cost. Manufacturing operations in emerging economies have increasingly adopted predictive analytics and localized inventory automation to address challenges related to supplier reliability, demand variability, and infrastructure limitations that are common in these market. Large multinational manufacturers operating in emerging markets have implemented sophisticated demand sensing systems that integrate multiple data sources including point-of-sale data, social media, sentiment, economic indicators to create more accurate demand forecasts for local markets.

Localized inventory automation complements predictive analytics by enabling distributed supply chain architectures that reduce dependence on centralized distribution models and improve responsiveness to local market conditions.
This approach involves implementing automated inventory management systems at multiple locations throughout the supply chain, enabling real-time adjustment of inventory levels based on predictive insights and local demand patterns. The combination of these technologies creates supply chain systems that are both intelligent and adaptive, capable of maintaining performance levels despite external disruptions and market volatility.

Supply chain resilience has become a critical imperative for businesses operating in emerging economies, where volatility, infrastructure limitations, and resource constraints create unique challenges for traditional supply chain management approaches. This review examines the integration of predictive analytics and localized inventory automation as strategic solutions for enhancing supply chain resilience in these dynamic markets. The research synthesizes current methodologies, implementation frameworks, and performance outcomes across various sectors in emerging economies. Predictive analytics approaches, particularly machine learning algorithms, time series forecasting, and demand sensing technologies, have demonstrated significant potential in addressing the complexity of supply chain environments where traditional reactive models fail to capture market volatility and disruption patterns. Emerging trends indicate growing adoption of hybrid models that combine predictive analytics with localized automation strategies, leading to more adaptive and responsive supply chain architectures. The findings suggest that while these technologies offer substantial improvements in supply chain performance, successful implementation requires careful consideration of local market conditions, regulatory environments, and stakeholder capabilities. Future research directions include developing more robust algorithms for data-scarce environments and addressing sustainability concerns in automated supply.

Several factors have contributed to the growing interest in these advanced supply chain technologies within emerging economies. The increasing availability of digital infrastructure, including mobile networks and cloud computing services, has made it feasible to implement sophisticated analytics and automation systems even in markets with limited traditional IT infrastructure. Additionally, the proliferation of e-commerce and digital payment systems in emerging markets has generated vast amounts of transactional data that can be leveraged for predictive analytics applications.

The automation component of localized inventory systems involves implementing intelligent systems that can make inventory decisions with minimal human intervention, using algorithms that incorporate real-time data, predictive insights, and business rules to optimize inventory levels continuously. These systems typically employ optimization algorithms, control systems, and decision support tools that can process large amounts of data and make rapid decisions in response to changing conditions. The automation enables faster response times and more consistent decision-making compared to manual inventory management processes.

Over the past few decades, managing inventory effectively is something companies have come to use to cut costs and still ensure adequate stock so as not to run into stock-outs in increasingly complicated supply chains. Traditional techniques used in inventory optimization cannot frequently achieve these multiple goals all at once, more especially with an increase in supply chain size and complexity. To address these difficulties effectively, researchers have applied some advanced approaches, including frameworks with simulation and multi-objective optimization methods. These methods combine complex computational tools with statistical techniques to attain a better balance of key inventory objectives.

Sustainability is an approach to address social, environmental, and economic challenges related to present business activities and needs. Moreover, sustainability is vital to enterprises business strategy to potentially reduce cost, minimize risks, and new opportunities that promotes social good. Thus, more enterprises give attention to sustainability as holistic approach, taking an account of the three pillars in all aspects of business. However, research regarding the applications of artificial intelligence for sustainability in SMEs and practices are scarce. Also, there is an increasing attention to apply long-term strategies to achieve SDGs, likewise, sustainability practices among SMEs. While there are sustainability efforts by SMEs, the applications of artificial intelligence and extent of practices that contributes to sustainability has been neglected which SMEs contribute significantly. Thus, it is relevant to study the applications of AI for sustainability present in the Philippines SMEs. This phenomenon is worth investigating using a qualitative approach to know the AI applications in business, SMEs sustainability progress and related issues. SMEs is evidently a major contributor of economic development in the Philippines, accounting 99.5% of all business establishments in the country. SMEs operates across industries including manufacturing, construction, retail, construction, and services, among others. To date, SMEs contributes to employment around 62.7% around of total employment. While SMEs have direct contribution to economy but these enterprises are challenged to outgrow competitors in the industry, through its innovativeness culture.

However, the development and deployment of AI

systems also raise significant challenges needing attention to ensure the responsible and ethical use of AI. Some of the most pressing challenges include the potential loss of jobs because of automation the lack of transparency in AI decision-making and the need for responsible data management. Additionally, there are concerns about the fairness and bias of AI systems, which can perpetuate existing social and economic inequalities. Also, the AI training and deployment may lead to high resource consumption.Likewise, lack of competence, integration of AI in sustainability goals, financial support, infrastructure, and cyberthreats are present challenges of using AI for business
to achieve sustainability.

Previous studies suggest the role of AI for SMEs
sustainability initiatives. SMEs have started automating
business processes, develop new business models, analyze
risks and safety, targets new customers, improve working
capital, material sourcing and logistics, and employee’s
productivity and behavior related footprint using AI. Likewise,
conversational AIs were used to facilitate ordering products,
obtain production information, and assist customer to product related feedbacks and concerns. In terms of project
management, AI is used to automate project management
activities such as estimation, resource management, KPIs,
prediction, experimentation, and security risks in construction. Likewise, AI is used to build recommender systems for adaptive source and inventory, human resource management and logistics and delivery using efficient routing and navigation systems. These AI applications in business
were found to contribute to achieving sustainability. On the other hand, the benefits of AI are numerous and far-reaching, and its impact is felt across business areas in
SMEs. Some recent findings found that (AI) is transforming
various SMEs, leading to improved efficiency, accuracy, and
better decision-making capabilities. Also, AI increased revenue by improving employees’ productivity through
streamlined business processes and efficient use of data. Moreover, AI in SMEs resulted in lower cost and improved
quality of products by understanding real-time product
controls.

Recently, SMEs in the Philippines have growing
appreciation of innovative technologies that supports
business and sustainability. For instance, mobile payments
have been adopted to provide efficient payment collection
system for SMEs involved retail and service activities.
Likewise, social media is evidently utilized in SMEs to target
customer segments and improve customer experience.
Moreso, SMEs were supported by startups with emerging

technologies that focuses on addressing problems on transport and logistics, zero-waste products, repurposing, energy-saving solutions, and locally sourcing of eco-friendly raw materials. Thus, SMEs in the Philippines continues to grow by utilizing technologies that results in economic prosperity and environmental sustainability. However, SMEs recognized the challenges dealing with large enterprises on sustainability. For instance, SMEs introduced sustainable practices and initiatives to change customers behavior to a greener lifestyle activity while addressing the market needs. Furthermore, SMEs have developed sustainable practices on greening logistics,information technology (IT) infrastructure, work practices, and waste management.More recently, SMEs considers artificial intelligence to add value to business operations and customers experience in many routine tasks. Hence, artificial intelligence can pave way to achieve sustainability in SMEs in the Philippines.

This study involves small and medium enterprises with evidence of utilization of artificial intelligence for business applications resulting in sustainability. The study initially reviewed a list of small and medium enterprises that have good innovation adoption rating and supported by different in select small and medium enterprises in the Philippines. Results show that a few SMEs have AI applications in business to address sustainability. While there are on-going sustainability efforts, most SMEs are in the incremental and situational development levels. This study confirms that insufficient physical and technological infrastructure, availability of data, customers privacy and security, insufficient legal frameworks, management support, and lack of AI adoption strategy are evident issues and challenges to progress AI for sustainability in SMEs. Thus, this needs much attention to enable more aligned sustainability efforts to take place among SMEs. However, the results of the study cannot be used to generalize all SMEs in the Philippines. As such, future work may focus on conducting a national study to provide comprehensive understanding of AI in SMEs. Likewise, cross-country studies may enrich understanding through comparative analysis of AI presence in SMEs of similar developing countries. On the other hand, AI is more likely be implemented in several business areas that persuade individuals to shift to greener lifestyle and behavior. Thus, this future gap can be further explored.

This review aims to provide a comprehensive analysis of current applications, methodologies, and outcomes associated with predictive analytics and localized inventory automation implementation in emerging economy supply chains. By examining both theoretical foundations and practical implementations, this work seeks to identify best practices, common challenges, and future research directions that will guide the continued development of resilient supply chain systems in developing markets.

CHAPTER 3
METHODOLOGY

3.1 Research Design

This study used a descriptive and developmental research design to create an Automated Resource Management System. The descriptive part was used to understand how the organization manages its resources find out what problems users face and figure out what the organization needs. This approach helped us get information about what the organization does what problems it has and what areas need to be improved.

The developmental part was about designing, building, testing and putting the proposed Automated Resource Management System into action. This approach allowed us to use the information we gathered and the requirements we found to create a working automated system that can manage resources make things more efficient and help with decision making for the Automated Resource Management System.

We used interviews, observations and surveys to get the information we needed. We did interviews to get ideas from users and people involved with the Automated Resource Management System about their experiences. What they expected. We watched how the organization works and how it manages its resources. We also gave out surveys to get feedback and make sure we understood what users needed for the Automated Resource Management System. The information we gathered was used to design and build the system.

The Software Development Life Cycle guided our study. It gave us a plan for planning, analyzing, designing, building, testing and putting the Automated Resource Management System into action. The Software Development Life Cycle made sure each step was done correctly and that the final system met the needs of its users. Our study was also based on Information Systems Theory, which says that technology, data, processes and people should work together to help the organization achieve its goals and improve how it manages resources for the Automated Resource Management System.

The way we did our research is important. It includes the methods, techniques, tools and processes we used to gather analyze and understand the information to achieve our goals for the Automated Resource Management System. Having a plan for our research made sure our findings about the Automated Resource Management System were valid, reliable and accurate. In this research the plan gave us a way to identify problems with resource management gather what users needed and build an automated solution, for the Automated Resource Management System. By following this approach we made sure the Automated Resource Management System we proposed was based on facts and could help the organization while making resource allocation, monitoring and use better.

3.2 System Development Methodology

The study used the Agile Development Model. This model is a way of working where people talk to each other and listen to users.The Agile Development Model is a repeating process. It involves people working together like developers and stakeholders.

Requirements Analysis: The first step was to figure out what the organization really needed.The study looked at problems with managing resources by hand. Researchers did interviews with people. Watched how things were done.

They also reviewed existing processes. Some problems were found, like not being able to get things done, having data issues taking a time to track resources. The Agile Development Model helped find these problems.

System Design: Next researchers made a plan for the system. They designed a database to store data in a way. A user-friendly interface was created.A plan for the system was also made.The System Design phase was crucial.It used the Agile Development Model. The goal was to make the system easy to use, able to grow and reliable.

Development: Then the system was built. The Development phase used the Agile Development Model. It was done in steps with each step improved. Users were asked for feedback to make the system better. The Agile Development Model helped minimize mistakes. It ensured the system and used throughout the Development phase.It really helped with managing resources. It was used to find problems and make the system better.

3.3 System Architecture

An Automated Resource Management System is a website that helps organizations manage their resources better. It is like a place where administrators and authorized users can see, assign and track resources in real time. This system makes resource management easier by automating tasks, reducing mistakes and making operations more efficient.
The system has important features such as It allows users to assign resources to tasks, departments or people and see what is available and being used. It also helps keep track of equipment, supplies and resources. The system controls who can see and use functions and information based on their role in the organization. The system also has a dashboard that shows summaries, statistics and pictures of how resources are being used.This helps management make decisions about planning and assigning resources.
With real-time data organizations can see trends, use resources better and be more productive.You can access the Automated Resource Management System through the internet using a browser.You do not need to install any software.This makes it easy to manage resources with an internet connection.The system also has a database that keeps information

up-to-date and available to authorized users. In short the Automated Resource Management System is an efficient way to manage resources.
It helps organizations use resources better work effectively and make better management decisions by automating resource management and providing accurate information.
The Automated Resource Management System makes it easy to manage resources and improve operations.

3.4 Tools and Technologies Used

3.5 Data Gathering Techniques

To get the information they needed for the Automated Resource Management System the researchers used a few ways to gather data.

Observation: This is where they watched how the company handles its resources now. They paid attention to how resources are written down, followed, given out and taken care of. This helped them see what is good and what is bad about the way things are done by hand. They saw where things are slow, where mistakes can happen and where things can go wrong when handling resources. By watching how things are done the researchers got an idea of how work really gets done and they used this to design a better automated system.

Interview: They talked to the people who handle resources like the people in charge, staff members and others who are involved. The researchers wanted to know what these people think about the system and what problems they have now. By talking to them the researchers got a good understanding of what the users need and want. This helped them figure out what the system needs to do and make sure it will solve the problems that exist now and handle resources.

To make sure the Automated Resource Management System works well we did a lot of testing while it was being developed. We wanted to find mistakes, check if the system does what it is supposed to do and make sure it meets the needs of the people who will use it.

3.6 Testing Procedure

Unit Testing: We tested each part of the Automated Resource Management System separately. This means we looked at things like registering resources, tracking inventory, managing users and generating reports. We tested each of these features on its own to make sure it works correctly. This helped us find and fix problems on which made the whole system better and more stable.

System Testing: After we put all the parts of the Automated Resource Management System together we tested the thing. We wanted to see if all the parts work well together and if the system does what it is supposed to do when it is being used normally. We tested a lot of things like how the different parts interact with each other, if the data is accurate, if the system is secure and how well the whole system performs.

User Acceptance Testing (UAT): We also had the people who will actually use the Automated Resource Management System test it. This included administrators and staff members. They looked at how the system works, how easy it is to use and if it really helps manage resources. We used their feedback to find things that need to be improved and to make sure the system meets their needs. When this testing was finished we knew the Automated Resource Management System was ready to be used.

3.7 Evaluation Criteria

To figure out if the Automated Resource Management System is any good we need to look at a things. We want to know if the Automated Resource Management System actually does what people need it to do and if it does it right.

Functionality: The Automated Resource Management Systems functionality is about whether it can do everything it is supposed to do. We will check if the Automated Resource Management System can manage resources, track inventory, handle users and make reports correctly. We also want to make sure that what the Automated Resource Management System produces is accurate and complete.

Usability: Using the Automated Resource Management System should be easy. We need to know if people can navigate the Automated Resource Management System without getting confused, understand what it can do and get their work done without any trouble. If the Automated Resource Management System is easy to use, people can work efficiently with training and they will be happy.

Reliability: The Automated Resource Management System needs to work all the time. We will see if the Automated Resource Management System can work without any errors, crashes or losing information. The Automated Resource Management System should work well all the time. Give us accurate results.

Security: The Automated Resource Management System needs to keep our data and resources safe from people who should not have access to them. We will look at how well the Automated Resource Management System checks who is using it, controls access, keeps data private and handles security measures. Security is very important because it keeps our information safe.

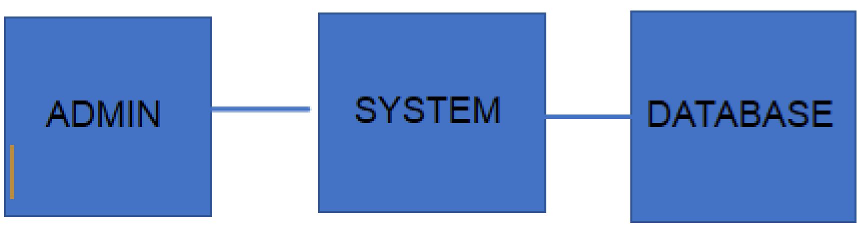
Efficiency: The Automated Resource Management System needs to use resources in a way that gets tasks done quickly. We will look at how fast the Automated Resource Management System responds, how quickly it processes information and how well it performs overall. If the

Automated Resource Management System is efficient it gets things done quickly makes people more productive and helps manage resources

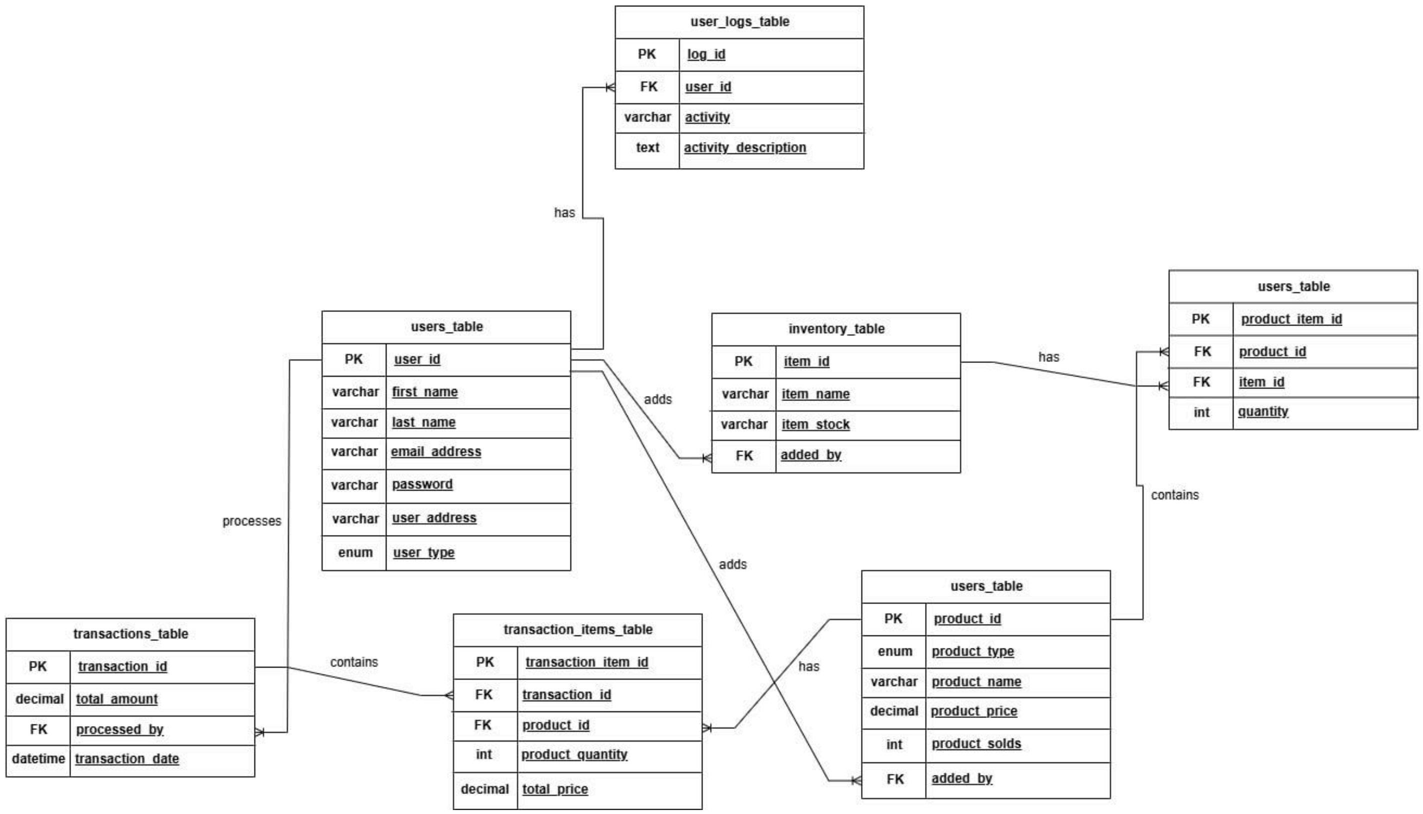
The results of these evaluation criteria will help us understand how well the Automated Resource Management System works, how good it is and if it is acceptable. The evaluation will also give us feedback to make the Automated Resource Management System better, in the future.

SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD – Level 0)



Entity Relationship Diagram (ERD)



Entity Attributes

- user_id (Primary Key)
- first_name
- last_name
- email_address
- password
- user_address
- user_type (Admin / Staff)
- log_id (Primary Key)
- user_id (Foreign Key → Users.user_id)
- activity
- activity_description
- item_id (Primary Key)
- item_name
- item_stock
- added_by (Foreign Key → Users.user_id)
- product_id (Primary Key)
- product_type
- product_name
- product_price I'm
- product_solds
- added_by (Foreign Key → Users.user_id)
- product_id (Foreign Key → Products.product_id)

- item_id (Foreign Key → Inventory.item_id)
- quantity
- transaction_id (Primary Key)
- total_amount
- processed_by (Foreign Key → Users.user_id)
- transaction_date
- transaction_item_id (Primary Key)
- transaction_id (Foreign Key → Transactions.transaction_id)
- product_id (Foreign Key → Products.product_id)
- product_quantity
- total_price

Relationships:

- A User can register several Inventory Items, meaning one user may handle the input or management of multiple stock records.
- A User can create multiple Products, showing that products are maintained and managed by system users such as administrators or staff.
- A User can handle multiple Transactions, where each transaction represents a completed process or sale.
- A Product can be included in many Transaction Items, meaning it can be sold repeatedly in different transactions.
- A Product can be included in many Transaction Items, meaning it can be sold repeatedly in different transactions.
- A Transaction can include multiple Transaction Items, where each item represents a specific product involved in that transaction.
- Each Transaction is managed by one User, identifying the person responsible for processing it.

End of Proposal

Note to the instructors handling this subject

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in application development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical considerations, user safety, and data privacy.

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